

SEED STENCILS FOR DIFFUSION THROUGH MATERIAL EDGES TOO STEEP FOR THE GRID

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ABSTRACT. *[Stub, #45: the three regimes of a diffusion coefficient (smooth, a jump, an edge of width δ below the node spacing h); what standard stencils and the 2016 jump construction do through the sub-grid edge; the seeds and their $\delta \rightarrow 0$ limit; the 1-D and 2-D results, elliptic and parabolic, against the coefficient treatments in use.]*

1. INTRODUCTION

[Stub, #45: $u_t = \nabla \cdot (\alpha \nabla u)$ with a coefficient that is smooth, jumps, or changes over a width $\delta < h$; the knee of standard stencils at $h \approx \delta$; the 2016–2017 interface-aware RBF-FD construction and its role as the $\delta \rightarrow 0$ limit; the seeds, whose basis is continued through the edge by ODEs; what is measured, and against what.]

Contributions. *[Stub, #45: the contributions, in the words LITERATURE.md section 6 allows (#43).]*

1.1. Relation to prior work. *[Stub, #43 supplies this paragraph (LITERATURE.md section 6a); #45 places it.]*

Outline. *[Stub, #45: one sentence per section.]*

2. SETTING

2.1. The problem and its interface conditions. *[Stub, #45: the heat equation and its equilibrium; continuity of u and of the flux $\alpha \partial_n u$ across a jump; the smooth sub-grid edge of width δ and the three regimes; the norms used.]*

2.2. One dimension. *[Stub, #45: equispaced grids, the naive operator assembled as $D_x A D_x$ (dissertation eq. 76), BD4 with $\Delta t \propto h$, the media and test problems.]*

2.3. Two dimensions. *[Stub, #45: the x -periodic strip, scattered nodes with rows straddling each interface, Gaussian RBF-FD with polynomial augmentation (42 nodes and degree 5 in the bulk, 30 and degree 4 at interfaces and boundaries), Dirichlet rows, the direct and iterative solvers.]*

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2.4. Interface-aware stencils for a jump. *[Stub, #45: the 2016–2017 construction as background: the translated polynomial basis, the continuity matrices with curvature, the warped RBFs.]*

2.5. Scope of the implementation. *[Stub, #45: the Python port and the published figures it reproduces (port notes); what it leaves out (the cornered interface), stated as scope.]*

3. SEEDS FOR THE DIFFUSION OPERATOR IN ONE DIMENSION

3.1. Seeds as the profiles of solutions polynomial in time. *[Stub, #46: the chain of seeds anchored at the evaluation node, continued through the edge in first-order form so that α is never differentiated; the seed-basis figure.]*

3.2. Nested kernels and the jump as the limit. *[Stub, #46: the nested kernels of the operator; the translated basis of Section 2.4 as the $\delta \rightarrow 0$ limit.]*

3.3. Well-posedness. *[Stub, #46: the seeds form an extended complete Chebyshev system, so every stencil solve is nonsingular.]*

3.4. Truncation error. *[Stub, #46: third order locally on the seeded rows, fourth globally; the two readings of the bound.]*

3.5. The elliptic problem in one dimension. *[Stub, #46: the exactness remark and its finite-volume twin, citing the literature #43 finds.]*

4. ONE-DIMENSIONAL RESULTS

4.1. Media and references. *[Stub, #47: the smooth-edged media and the references at every δ .]*

4.2. The parabolic knee. *[Stub, #47: the naive knee at $h \approx \delta$; the snapshot of the four regimes on one grid.]*

4.3. The jump construction on a smooth edge. *[Stub, #47: the $\delta = 0$ construction's error on a smooth edge.]*

4.4. The seeds. *[Stub, #47: the seeds' rates at every δ .]*

4.5. Coefficient treatments. *[Stub, #47: the treatments' table and the ranking; the elliptic sanity figure as a remark.]*

5. SEEDS IN TWO DIMENSIONS

5.1. A straight feature. *[Stub, #48: the local frame and the anchor; seeds polynomial in the tangential coordinate with coefficient functions marched along the normal.]*

5.2. Stencil assembly. *[Stub, #48: the seeds in the RBF-FD system, the warp they bring with them, which rows are seeded; the seed-section figure.]*

5.3. Curved and tangentially varying features. *[Stub, #48: why a frozen normal profile is inconsistent where α varies along the edge; the tangential chain in the curve's own coordinates.]*

5.4. A thin resistive layer. *[Stub, #48: the double edge of a ring: its exact width, series composition, the flux seeds, the warp on rows anchored inside the layer.]*

6. TWO-DIMENSIONAL RESULTS

6.1. Test problems and references. *[Stub, #49: the flat band, the sine pair, the ring; the separable, product-grid and reference-free references (stiff 4.1, 4.6, 4.8; statement 1).]*

6.2. Naive baseline. *[Stub, #49: the knee on scattered nodes and the jump construction on its floor (stiff 4.2; statements 2–3).]*

6.3. Solvability and spectra. *[Stub, #49: the seed rows in the global matrix (stiff 4.4; statement 6).]*

6.4. The flat sweep. *[Stub, #49: the seeds against the baselines at every (n, δ) , the warp, the elliptic ranking (stiff 4.5; statements 4, 5, 7).]*

6.5. The curved feature. *[Stub, #49: the frozen profile and the tangential chain on the sine pair (stiff 4.6–4.7; statement 8).]*

6.6. The ring and the extremizing sweep. *[Stub, #49: the twins of EABE Fig. 19–20 and the smooth ring (stiff 4.8, 4.10; statement 9).]*

6.7. Coefficient treatments. *[Stub, #49: the treatments on scattered nodes (stiff 4.9; statement 10).]*

6.8. A snapshot. *[Stub, #49: the four regimes on one node set (stiff 5.1; statement 11).]*

7. DISCUSSION

7.1. Limitations. *[Stub, #50: corners, extreme contrasts, the cost in three dimensions, what the experiments leave open.]*

7.2. Future work. *[Stub, #50: in the order to do it.]*

7.3. Conclusions. *[Stub, #50: LITERATURE.md section 6b, verbatim.]*

ACKNOWLEDGMENTS

[Stub, #52: acknowledgments; where the code and data live.]

TOOL AND COMPUTATIONAL RESOURCE DISCLOSURE

[Stub, #52.]

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